

5. Ratemaking in MTPL line of business The general structure

Risk Premium

As well-known, in ratemaking non-life insurance policies, the first step is the assessment of the expected aggregate amount of claims at portfolio level as:

$$P = E(X)$$

where the r.v. X considers both payments and claims reserve of claims incurred in a specific accident year.

Usually this estimate is based on a separate valuation of average cost (AC) and frequency (f) of different accident years. We have indeed that the average risk premium is:

$$\bar{P} = E(f) E(AC)$$

Estimation of «basic» expected frequency and average cost

Usually, the estimation of expected frequency and average cost is based on the empirical analysis of the portfolio over last years.

Number and costs of claims **already paid and expected to be paid in future are considered.**

<i>Calendar Years</i>	<i>Average Cost</i>	<i>Frequency</i>
t-4	AC_{t-4}	f_{t-4}
t-3	AC_{t-3}	f_{t-3}
t-2	AC_{t-2}	f_{t-2}
t-1	AC_{t-1}	f_{t-1}
t	AC_t	f_t

An Example

Calendar Year	Exposures (Vehicles)	Number of Claims	Amount of Paid Claims	Estimated Amount of Open Claims	Total Amount	AC	f
t-5	124978	11571	18,198,206.00	28,790,109.00	46,988,315.00	4,060.87	9.3%
t-4	134871	12271	18,425,206.00	29,108,874.00	47,534,080.00	3,873.69	9.1%
t-3	154378	13201	18,952,206.00	29,208,874.00	48,161,080.00	3,648.29	8.6%
t-2	165871	13922	29,734,804.00	23,468,919.00	53,203,723.00	3,821.56	8.4%
t-1	178373	14805	28,215,916.00	24,702,042.31	52,917,958.31	3,574.33	8.3%

	$\widehat{AC}_t$	$\hat{f}_t$
Selected	3,756.00	8.30%

Table 1 – Average cost of claims and claims frequency in the motor and marine liability insurance sector
Values in €

PANEL A: Excludes claims IBNR, contribution to the Road Accident Victims Guarantee Fund and other residual items								
Year	Claims frequency %	Change %	Average claim cost - property damage	Change %	Average claim cost - personal injury	Change %	Average total claim cost**	Change %
2000	9.82%	-1.3%	1,278	2.9%	9,920	14.9%	2,809	13.1%
2001	8.54%	-13.1%	1,431	12.0%	11,175	12.7%	3,186	13.4%
2002	7.82%	-8.4%	1,535	7.3%	12,686	13.5%	3,532	10.9%
2003	7.66%	-2.1%	1,634	6.4%	13,542	6.7%	3,805	7.7%
2004	7.61%	-0.6%	1,701	4.1%	13,206	-2.5%	3,982	4.7%
2005	7.55%	-0.8%	1,644	-3.3%	13,106	-0.8%	4,047	1.6%
2006	7.47%	-1.1%	1,674	1.8%	13,233	1.0%	4,100	1.3%
2007	7.61%	1.9%	1,764	5.4%	11,958	-9.6%	3,967	-3.2%
2008	7.73%	1.6%	1,772	0.5%	11,830	-1.1%	3,913	-1.4%
2009	7.77%	0.5%	1,725	-2.7%	11,694	-1.1%	3,903	-0.3%
2010	7.36%	-5.2%	1,716	-0.5%	12,052	3.1%	4,057	4.0%
2011	6.53%	-11.3%	1,803	5.0%	13,155	9.2%	4,345	7.1%
2012	5.87%	-10.1%	1,899	5.3%	14,804	12.5%	4,495	3.5%
2013	5.65%	-3.8%	1,883	-0.8%	15,986	8.0%	4,564	1.5%
2014	5.48%	-2.9%	1,894	0.6%	16,150	1.0%	4,532	-0.7%
2015	5.55%	1.2%	1,908	0.7%	16,389	1.5%	4,467	-1.5%
2016	5.65%	1.8%	1,912	0.2%	16,132	-1.6%	4,374	-2.1%
2017	5.61%	-0.7%	1,941	1.5%	16,297	1.0%	4,326	-1.1%
2018*	5.43%	-3.2%	1,977	1.9%	16,514	1.3%	4,361	0.8%

(*) ANIA estimates based on advance information on 2017 financial statements

(**) Source: IVASS; for 2018, data from supervisory reporting forms

Claims frequency is defined as the ratio between

- the number of claims incurred and reported during the accident year that have given (or will give rise) to compensation
- the number of vehicles exposed to the risk of claim-generating accidents (measured on the basis of days of exposure during the year, converted into “vehicle-years”).

- **The average cost of claims** is derived by dividing the total cost of claims (paid and reserved) by their number.
 - It takes account both of final and partial payments and future expected payments for open claims that have been reported (reserved amounts).
 - It excludes claims incurred but not reported (IBNR reserves), contributions to the Road Accident Victims Guarantee Fund and other residual items.

Forecasting final Average cost

Basic average cost $\hat{AC}_t$ must be corrected to consider several aspects

- future development of open claim counts, future payments for open claims, IBNR claims;
- period of coverage of the new tariff (moving from accident year t to accident year $t + 2$);
- specific claims handling expenses (“Spese di resistenza”)

Typically, three multiplicative coefficients c_i^{AC} ($i = 1, 2, 3$) are computed such that:

$$\hat{AC} = \hat{AC}_t \cdot \prod_{i=1}^3 c_i^{AC}$$

Development of open claims

In order to consider future development of open claim counts, we estimate the proportion $\hat{p}_{re}$ of future re-opened and/or closed without payments claims.

This percentage represents an estimation of the difference of future re-opened claims and closed without payments divided by the sum of number of closed claims with payments and number of open claims at the valuation date.

By using this percentage, we can estimate the new proportion of paid and open claims as:

$$\hat{p}^R = \frac{\frac{n^R}{n^R + n^P} + \hat{p}_{re}}{1 + \hat{p}_{re}}$$

$$\hat{p}^P = 1 - \hat{p}^R$$

where n^P and n^R are number of payments and number of open claims, respectively.

Average cost of open claims

Dev. Year	$\hat{AC}_{t,i}$	Cumulative settl. speed	Incremental settl. speed	Normalized Incremental settl. speed	$\hat{C}_{t,i}$
0	$\hat{AC}_{t,0}$	s_0^C			
1	$\hat{AC}_{t,1}$	s_1^C	s_1^I	$s_1^{N,I}$	$\hat{C}_{t,1}$
2	$\hat{AC}_{t,2}$	s_2^C	s_2^I	$s_2^{N,I}$	$\hat{C}_{t,2}$
—	—	—	—	—	—
T	$\hat{AC}_{t,T}$	$s_T^C = 1$	s_T^I	$s_T^{N,I}$	$\hat{C}_{t,T}$
Total			$\sum_{i=1}^T s_i^I$	$\sum_{i=1}^T s_i^{N,I} = \hat{p}^R$	$\hat{C}_t = \sum_{i=1}^T \hat{C}_{t,i}$

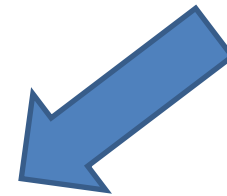
where:

$$s_i^{N,I} = s_i^I \frac{\hat{p}^R}{1 - s_0^C}$$

$$\hat{C}_{t,i} = \hat{AC}_{t,i} \cdot s_i^{N,I}$$

So we can estimate a new average cost for open claims as:

$$\hat{AC}_t^R = \frac{\hat{C}_t}{\hat{p}^R} = \frac{\sum_{i=1}^T \hat{C}_{t,i}}{\hat{p}^R} = \frac{\sum_{i=1}^T \hat{AC}_{t,i} \cdot s_i^{N,I}}{\hat{p}^R}$$



Coefficient $\hat{c}_1$

Hence, we have:

$$\hat{c}_1^{AC} = \frac{\hat{p}^P \cdot AC_t^P + \hat{p}^R \cdot \hat{AC}_t^R}{\hat{AC}_t}$$

where

- AC_t^P average cost for paid claims
- $\hat{AC}_t^R$ estimated average cost for open claims
- the coefficient $c_r = \frac{\hat{AC}_t^R}{AC_t^R}$ measures the effect of the new estimation of average cost for open claims.

Coverage period

- The aim of coefficient $\hat{c}_2^{AC}$ is to estimate the aggregate claim amount during the coverage period. It catches changes in average cost from calendar year t to the coverage period.
- In particular, for annual policies, we are interested in the period that goes from the beginning of the product life-cycle (market introduction stage) to the product termination plus one-year.
- We aim at estimating proportion of claims ϕ_{t+h} (with $\sum_h \phi_{t+h} = 1$) incurred in each calendar year covered.

For instance, if annual policies are sold from 1st of July $t + 2$ to end of June $t + 3$ and we assume uniform distributions of premiums and claims, we have: $\hat{\phi}_{t+2} = 0.25, \hat{\phi}_{t+3} = 0.5, \hat{\phi}_{t+4} = 0.25$

Average cost of open claims

1. To estimate future average costs and settlement speeds for each period $t + h$ (in Table we assume $h = 2, \dots, 4$).

Accident Year	Development Year			
	0	1	—	T
t	$\hat{C}_{t,0}$	$\hat{C}_{t,1}$		$\hat{C}_{t,T}$
—	—	—		—
t+2	$\hat{C}_{t+2,0}$	$\hat{C}_{t+2,1}$		$\hat{C}_{t+2,T}$
t+3	$\hat{C}_{t+3,0}$	$\hat{C}_{t+3,1}$		$\hat{C}_{t+3,T}$
t+4	$\hat{C}_{t+4,0}$	$\hat{C}_{t+4,1}$		$\hat{C}_{t+4,T}$

2. To estimate future average cost for paid claims as

$$\hat{AC}^P = \frac{\sum_{h=2}^4 \hat{C}_{t+h,0} \cdot \hat{\phi}_{t+h}}{\sum_{h=2}^4 \hat{p}_{t+h}^P \cdot \hat{\phi}_{t+h}} = \frac{\sum_{h=2}^4 \hat{AC}_{t+h,0} \cdot \hat{p}_{t+h}^P \cdot \hat{\phi}_{t+h}}{\sum_{h=2}^4 \hat{p}_{t+h}^P \cdot \hat{\phi}_{t+h}}$$

3. To estimate future average cost for open claims as

$$\hat{AC}^R = \frac{\sum_{j=1}^T \sum_{h=2}^4 \hat{C}_{t+h,j} \cdot \hat{\phi}_{t+h}}{\sum_{h=2}^4 \hat{p}_{t+h}^R \cdot \hat{\phi}_{t+h}} = \frac{\sum_{j=1}^T \sum_{h=2}^4 \hat{AC}_{t+h,j} \cdot \hat{s}_{t+h,j}^I \cdot \hat{\phi}_{t+h}}{\sum_{h=2}^4 \hat{p}_{t+h}^R \cdot \hat{\phi}_{t+h}}$$

Coefficient $\hat{c}_2$

Hence, we have:

$$\hat{c}_2^{AC} = \frac{\hat{p}^{P,\star} \cdot \hat{AC}^P + \hat{p}^{R,\star} \cdot \hat{AC}^R}{\hat{p}^P \cdot AC_t^P + \hat{p}^R \cdot AC_t^R}$$

where

$$- \hat{p}^{P,\star} = \sum_{h=2}^4 \hat{p}_{t+h}^P \hat{\phi}_{t+h}$$

$$- \hat{p}^{R,\star} = \sum_{h=2}^4 \hat{p}_{t+h}^R \hat{\phi}_{t+h}$$

Forecasting final frequency

Similarly to average cost, basic frequency $\hat{f}_t$ must be corrected to consider several aspects

- future development of open claim counts (re-opened and closure) and IBNR claims
- period of coverage of the new tariff (moving from accident year t to accident year $t + 2$).

Typically, two multiplicative coefficients c_i^f ($i = 1, 2$) are computed such that:

$$\hat{f} = \hat{f}_t \cdot \prod_{i=1}^2 c_i^f$$

Pure Premium

- Average pure premium is derived as:

$$\hat{P}^P = \hat{f} \cdot \hat{AC} \cdot c^{fin}$$

- Investment incomes on the reserves are partially recognized in premium assessment via a specific coefficient c^{fin} .

The coefficient is typically computed as the ratio of the discounted average cost to the undiscounted one $\hat{AC}$.

Discounted average cost is evaluated by assuming the expected rate of return and by considering expected timing of future cash-flows.

- Average gross premium is:

$$\hat{P}^T = \frac{\hat{P}^P}{1 - \alpha - \beta}$$

with

- α : loading coefficient (expenses and safety).
- β : coefficient for FVGS contribute. The Italian guarantee fund for victims of hit-and-run drivers, is called “Fondo Garanzia Vittime della Strada” (FGVS) and it is financed by a fixed percentage loading applied to the pure premium net of administration costs. This percentage loading is actually 2.38% of gross premiums.

<i>Average Gross Premium</i>	
<i>Sector I –Motor Vehicle</i>	
(1) Basic Average Cost	3.756,00
<i>Coefficients</i>	
(2) Effect of development of payments for open claims and IBNR	1,0200
(3) Effect of Coverage period (from ay t to ay t+2)	1,1500
(4) Claims handling expenses (Spese di resistenza)	1,0000
(5) Final Average Cost (Tariff AC) (5) = (1)x(2)x(3)x(4)	4.405,79
(6) Effect of financial rate of return of Claims Provisions	0,9315
(7) Basic Frequency	0,083
<i>Coefficients</i>	
(8) IBNR and re-opened claims	1,0934
(9) Effect of Coverage period (from ay t to ay t+2)	0,98
(10) Final Frequency (Tariff f)(10) = (7)x(8)x(9)	0,0889
(11) Average Pure Premium (11) = (5)x(6)x(10)	365,00
(12) Contribute to FVGS	11,74
(13) Loadings (expenses and safety)	110,55
(14) Average Gross Premium (14) = (11)+(12)+(13)	487,29

Average Gross Premium (net of taxes and national health service)

Specific premium rates

Risk classification and experience rating

- We dealt with premium calculation assuming that the same premium rate is applied to each policy. This is justified when policies are similar, i.e. homogeneous (except possibly for the sum insured and the time of issue or renewal).
- However, policies always differ for some features; as well-known such differences suggest the adoption, within the same line of business, of specific premium rates
- Policies for which the insurer can assume the same attitude to record claims are usually grouped into a risk class. The definition of a risk class is based on:
 - risk factors, i.e. the features of a risk which prove to explain the claim experience (in the example above, the risk factors are: use, location, building materials and number of floors);
 - the outcomes (or modes) of each risk factor, which can be either qualitative or quantitative.

Specific premium rates

Risk classification and experience rating

In order to assess a customized premium that considers the risk-profile of the specific policyholder, some predictors (i.e. the rating variables) are used. For instance, assuming only three rating variables, we have that the risk premium of a specific policyholder, belonging to the rating class i, j, h , is derived as:

$$P_{i,j,h} = P_b \alpha_i \beta_j \gamma_h \quad (1)$$

with a basic rate P_b and rate relativities $\alpha = [\alpha_i]$, $\beta = [\beta_j]$ and $\gamma = [\gamma_h]$ used to assess premiums for all non-base levels.

The base rate P_b is typically altered in order to assure the equilibrium at portfolio level:

$$\sum_{i,j,h} P_b \alpha_i \beta_j \gamma_h \cdot v_{i,j,h} = P \cdot v$$

with $v_{i,j,h}$ the number of policies belonging to the class i, j, h and $v = \sum_{i,j,h} v_{i,j,h}$
Hence, we have:

$$P_b = \sum_{i,j,h} \alpha_i \beta_j \gamma_h \cdot v_{i,j,h} = P \frac{v}{\sum_{i,j,h} \alpha_i \beta_j \gamma_h \cdot v_{i,j,h}} \approx \frac{P}{\bar{\alpha} \cdot \bar{\beta} \cdot \bar{\gamma}} \quad \leftarrow$$

Other elements

- Law 40/2007 (known as “Bersani-bis”), in effect since 3 April 2007, introduced two mandatory provisions concerning bonus-malus motor insurance policies:
 - the obligation for the insurer to apply to the policy on an additional vehicle acquired by the owner of one already insured or by a permanent component of the owner’s household the same merit class as that assigned to the owner of the currently insured vehicle;
 - the obligation for the insurer to apply merit penalties (“malus”) only when the policy-holder bears the “primary liability” for an accident, thereby excluding cases of minor liability and equal joint liability (50% each)
- Both measures, in view of the unwarranted benefits obtained by the persons involved (new policyholders and drivers responsible for accidents) have redistributed the insurers’ premium income requirement;
- **This law helped to hasten the crisis of the system**, which is now incapable of performing its original function, namely correlating the insured’s premium with his driving record over.
- It is noteworthy that the law has been also extended with application since 16 of February 2020 of the so-called “Bonus-Malus familiare” with the modification of the art. 134 of Insurance Code (CAP).
- Given this situation, the insurer aims at estimating a **specific “Bonus-Malus” multiplicative coefficient that considers the effect on total premiums of expected policyholders’ movements between different classes (to “creep” towards top classes)**. The diminution in insurers’ expected premium income due to the price reductions imposed in the types of contract covered by the law result in a disequilibrium that has to be made good **by redistributing the income shortfall over the entire set of policyholders.**
- A second coefficient is related **to the rate discounts provided by the distribution channel**. On the basis of the predetermined set of discounts (that the distribution channel has at disposal), it is estimated the average increase of basic premium to reach the equilibrium at portfolio level.

Basic Gross Premium	
Sector I –Motor Vehicle	
(14) Average Gross Premium	487,29
Effect of Bonus-Malus on policyholder distribution	1,0100
Gradients:	
Location	0,6874
Vehicle Power	2,2589
Coverage level (retention)	1,0826
Bonus-Malus	0,5986
Age/Gender	0,9878
Job	0,8975
Other Vehicle characteristics	0,9841
Insurance premiums in installments (monthly or semi-annual)	1,0587
Product of gradients	0,9295
Gross Premium	529,52
Theoretical coefficient for distribution channel discounts	1,1478
Final Gross Premium (net SSN and taxes)	607,78

Further development and current practice

- Use of telematics to reduce fraud and to refine customized premiums (i.e. adding new variables to or replacing traditional rating variables)
- Diffusion of Usage-Based Insurance (UBI) solutions
 - Pay How You Drive» (PHYD)
 - Pay as you drive» (PAYD)
- Possible elements that affect the risk
 - Driving Behaviour
 - Environment
 - Exposures based on km
 - Other external data (weather, traffic, other)